

HENGJIN LIU

South China University of Technology
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EDUCATION

South China University of Technology

Guangzhou, China

Supervisor: Prof. Zhaoqing Feng

M.Sc. in Theoretical Physics

Sep 2020 – July 2023

Heilongjiang University of Science and Technology

Heilongjiang, China

B.Eng. in Safety Engineering

Sep 2015 – July 2019

PUBLICATIONS

[Physical Review C]	Collective flows of clusters and pions in heavy-ion collisions at GeV energies (In English) Heng-Jin Liu, Hui-Gan Cheng, Zhao-Qing Feng Physical Review C, IF: 3.4, 2023. doi: https://doi.org/10.1103/PhysRevC.108.024614
[Science China: Physics, Mechanics & Astronomy]	Nuclear Medium Effects and Collective Flow of Pion Production in Intermediate-Energy Heavy-Ion Collisions (In Chinese) Heng-Jin Liu, Zhao-Qing Feng Science China: Physics, Mechanics & Astronomy, IF: 1.087, 2023. doi: 10.1360/SSPMA-2022-0473
[Nuclear Technology]	Recent Advances in the Production of Strange Particles and Hypernuclei in Intermediate- and High-Energy Heavy-Ion Collisions (In Chinese) Zhao-Qing Feng, Heng-Jin Liu, Hui-Gan Cheng, Si-Na Wei Nuclear Technology, IF: 0.795, 2023. doi: 10.11889/j.0253-3219.2023.hjs.46.080010
[Submitted to Nuclear Physics A.]	Shedding light on the pion production in heavy-ion collisions and application into the neutron star matter properties (In English) Heng-Jin Liu, Ban Zhang, Yuan-Qing Guo, Hui-Gan Cheng, Si-Na Wei, Zhao-Qing Feng. arXiv:2302.02131 [nucl-th]

RESEARCH EXPERIENCE

Theoretical Physics Group, South China University of Technology

Guangzhou, China

Research assistant, supervised by Prof. Zhao-Qing Feng

Sep 2020 – Jun 2023

Project: Pion Production Dynamics & High-Density Symmetry Energy Constraints (Based on LQMD Model)

Key Contributions:

- Implemented medium modifications to $\Delta(1232)$ resonance production cross-sections, introduced Δ -nucleon interaction potential, pion optical potential, and resonance decay energy conservation corrections.
- Simulated 7 reaction systems ($^{40}\text{Ca}+^{40}\text{Ca}$, $^{90}\text{Zr}+^{90}\text{Zr}$, $^{96}\text{Zr}+^{96}\text{Zr}$, $^{96}\text{Ru}+^{96}\text{Ru}$, $^{197}\text{Au}+^{197}\text{Au}$, $^{132}\text{Sn}+^{124}\text{Sn}$, $^{108}\text{Sn}+^{112}\text{Sn}$) to analyze pion yields, rapidity/transverse momentum spectra, and medium effect mechanisms.
- Studied collective flows (V_1 - V_4) of pions, protons, and clusters (deuteron, triton, ^3He , ^4He) via Wigner density method, achieving good agreement with FOPI/HADES experimental data.
- Core Achievement:** Constrained high-density symmetry energy using π^-/π^+ ratios and neutron-proton ratios, deriving slope parameter $L = 42 \pm 25 \text{ MeV}$ or stiffness parameter $\gamma_s = 0.3$.

Technical & Methodological Competence

- Proficient in Lanzhou Quantum Molecular Dynamics (LQMD) model and heavy-ion collision simulation.
- Expertise in analyzing nuclear reaction observables (particle yields, collective flows, momentum/rapidity spectra).

- Skilled in medium effect corrections (optical potential, cross-section modification) and energy conservation implementation.
- Experienced in data validation against international collaborations (FOPI, HADES, SπRIT).
- Competent in nuclear physics mathematical derivation (e.g., Clebsch-Gordan coefficients) and statistical data analysis.

WORKING EXPERIENCE

BYD Auto Industry Co., Ltd. (Shenzhen, China)

Senior Vehicle Safety Engineer

Aug 23-Dec 25

Project 1: Research on Occupant Diversity Protection in Frontal Impacts. Analyzed injury sensitivity across crash pulses, cabin dimensions, and restraint system parameters.

- Utilized Hybrid III & THOR-50M finite element dummies and response surface methodology (RSM) via simulation.
- Established relationships between head/chest injury metrics and key parameters.

Outcome: Authored CNCAP2027 & CIASI2026 Safety Development Guideline – Frontal Occupant Diversity, adopted company-wide; delivered annual FEA training for new hires.

Project 2: Study on Seat Belt Anchor Position Effects on Routing Path and Hybrid III 5th Thoracic Compression.

- Resolved Stage S3 excessive chest compression via analysis of crash pulses, dummy positioning and pelvis acceleration., etc.
- Conducted optimal seatbelt routing analysis to minimize chest injury metrics for female dummies. Established a mathematical correlation between upper anchorage positioning and seatbelt routing by numerical regression, deriving a predictive formula to guide vehicle development projects.

Outcome: Conference paper selected for 31st China SAE Conference Proceedings.

SELECTED AWARDS AND HONORS

- First-prize Scholarship of SCUT 2022
- First-prize Scholarship of SCUT 2021
- First-prize Scholarship of SCUT 2020
- Excellent Case Award from Full-Vehicle Technology Integration Development Center: "Case Study Report on Abnormal PAB Deployment and IP Tearing Issues"
- Second Prize for Outstanding Technical Project Award: "Mechanism Study of Active-Passive Safety Systems Aligned with China NCAP Protocol Upgrades"

INVITED TALKS AND MEDIA COVERAGE

Invited Talks and Presentations

Hengjin Liu, "Study of Collective Flow and Symmetry Energy in Mid-energy Heavy Ion Collisions",
Guangdong-Hong Kong-Macao Nuclear Physics Forum, [2023]

SERVICES

Club

- President of the Skateboard Club Sep 2017 - Sep2018

Teaching

- University Physics (TA) Sep 2021 - Sep2022